

Dealing with Uncertainty

Chapter 9 — Forecasting & Decision Making

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Learning Objectives

By the end of this chapter, you will be able to:

- **Distinguish** between Risk and Uncertainty in innovation.
- **Apply** forecasting techniques to reduce uncertainty (Delphi, Scenarios).
- **Evaluate** the political and cognitive nature of decision-making.
- **Understand** the concept of the "Funnel" in selecting projects.
- **Identify** cognitive biases that lead to poor innovation choices.

The Selection Challenge

Innovation is a lottery. We start with many ideas, but we have limited resources.

The Core Problem

How do we choose which options to fund when the future is unknown?

- If we are too conservative → We miss the next big thing (Type II Error).
- If we are too risky → We waste money on failures (Type I Error).

The Goal: To move from "Gambling" to "Managed Risk."

Risk vs. Uncertainty (Knight)

They are not the same thing.

Risk

- We know the possible outcomes.
- We can calculate the probability of each.
- Example: Rolling a dice or actuarial tables.
- **Manageable via Calculation.**

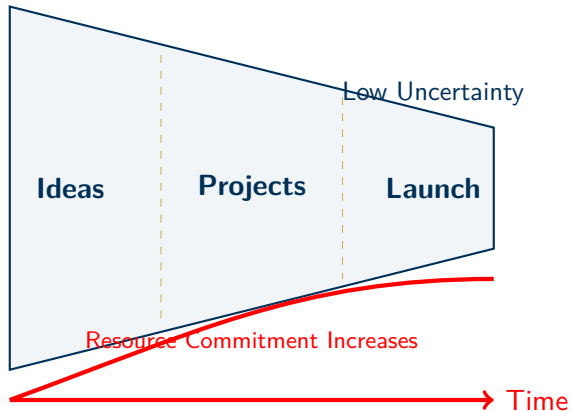
Uncertainty

- We do not know the outcomes.
- Probabilities are impossible to calculate.
- Example: Launching a radical new technology.
- **Manageable via Learning/Forecasting.**

The Selection Funnel

The purpose of the process is to reduce uncertainty over time.

High Uncertainty



Forecasting: Looking into the Fog

Since we cannot calculate probabilities for radical innovation, we use forecasting.

Type	Methods
Quantitative	Trend Extrapolation, Regression Analysis, S-Curves. (Works best for <i>incremental</i> change).
Qualitative	Benchmarking, Brainstorming, Delphi Method, Scenario Planning. (Works best for <i>radical</i> change).

The Delphi Method

Structured consensus building among experts.

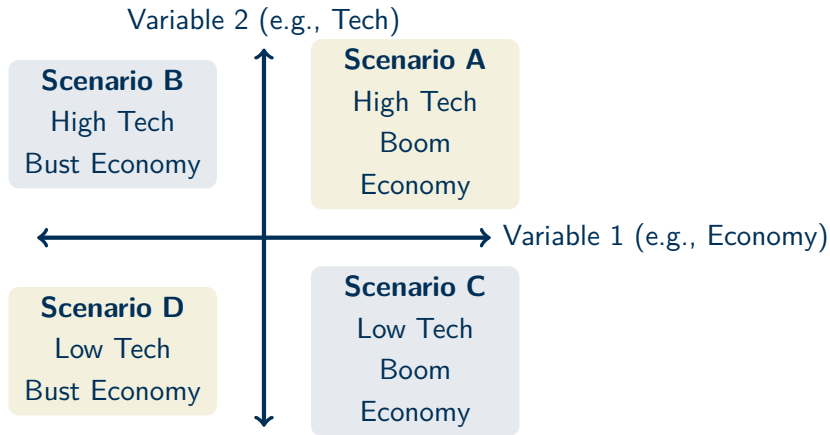
1. **Selection:** Choose a panel of experts (diverse backgrounds).
2. **Round 1:** Experts answer a questionnaire anonymously.
3. **Aggregation:** Moderator summarizes results and feedback.
4. **Round 2:** Experts revise their answers based on the group's feedback.
5. **Consensus:** Repeat until the range of answers narrows.

Benefit

Removes the "loudest voice in the room" bias by maintaining anonymity.

Scenario Planning

Instead of predicting *one* future, imagine *multiple* futures.



We test our strategy against ALL four scenarios to ensure resilience.

Once we have data/forecasts, how do we decide?

- **Financial Methods:** Payback period, NPV, ROI.
 - *Flaw:* Often kills radical innovation because numbers look bad early on.
- **Portfolio Matrices:** BCG Matrix, Risk/Reward Matrix.
 - *Goal:* Balance "Cash Cows" with "Wild Cats."
- **Checklists/Scoring Models:** Multi-criteria decision making (e.g., Strategic Fit + Tech Feasibility + Market Potential).

Selection is not just analytical; it is political. Innovators must build a coalition.

The Coalition of the Willing

To survive the selection process, an idea needs:

- **Power:** Sponsorship from senior leaders.
- **Resources:** Budget and talent.
- **Proof:** Prototypes or customer data (Minimum Viable Product).

"Data wins arguments, but stories win funding."

Why We Make Bad Decisions

Managers are human, and humans are irrational.

Common Traps

- **Confirmation Bias:** Seeking only data that supports our idea.
- **Sunk Cost Fallacy:** "We've spent so much, we can't stop now."
- **Escalation of Commitment:** Doubling down on a failing project.
- **Not Invented Here:** Rejecting good ideas because they came from outside.

Solution: Diversity in decision-making teams and rigorous "Stage-Gate" reviews.

Summary:

1. Dealing with uncertainty is the core challenge of the **Selection** phase.
2. We must distinguish between calculable **Risk** and ambiguous **Uncertainty**.
3. Use **Delphi** and **Scenario Planning** to map the unknown.
4. Avoid relying solely on financial metrics for radical innovation.
5. Be aware of **Cognitive Biases** that distort decision-making.

Thank You

Questions?